

**Does Transient Attention Affect Unconscious Processing of Perceived
Contrasts?**

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Introduction

Attention being our filter towards an external visual scene, we have two important cognitive “tools” to deem information irrelevant that would otherwise impose on us an immense cognitive load. Either we direct our attention by orienting our gaze (overt attention), or we shift attention without any eye movements, labelled as covert attention (Carrasco et al., 2004). The latter can be oriented at will or it can also be changed abruptly by an external stimulus, such as when driving a car, we notice in our periphery a rolling ball on the road (Carrasco et al., 2004). Such a *transient attention* can improve our perception of the external world. This claim manifests most clearly in an experimental paradigm developed by Carrasco et al. (2004), where it was found that transient covert attention ameliorates the phenomenological appearance of perceived contrasts. The authors demonstrated that when a target stimulus is preceded by an uninformative cue, contrast of the target stimulus is subjectively perceived higher than objectively presented. Since then, many experimental paradigms have shown how attention modifies visual appearance, from increased contrast perception to increased spatial frequency, increased saturation or even change in object size (Carrasco & Barbot, 2019). Furthermore, Liu et al. (2005) illustrated how such effects have their neural correlates in the brain. It is yet unknown whether this altered phenomenological sensitivity to contrast perception is a conscious or unconscious process.

To this end, the current project has set up two goals. First, the aim was to have a direct replication of the experimental paradigm of Carrasco et al. (2004). For this purpose, I programmed the experimental procedure in PsychoPy, after which I recruited five participants to perform the task in LAB B-224, set up in the Biology building of the Umeå University. Second, we intended to modify the procedure so that the processing of the target stimulus becomes unconscious. Due to the limited timeframe of the project, this version of the experiment was only conducted on one participant. In summary, the direct replication was judged to be successful, while the unconscious version of the experiment uncovered some limitations and possible future directions that will be discussed at the end of the report.

An approximate timeline of the project follows. The first two weeks were allocated to program the experiment in PsychoPy, following the same steps as it was reported by Carrasco et al. (2004). By the end of the second week the task was ready, and I served as the first participant to perform it. On week three I conducted data analysis on my data that revealed a successful replication of the experimental paradigm. Hence, weeks four and five followed by gathering data from five more participants (although one of them had to be excluded due to not being able to perform the task).

During data collection, preparation for the unconscious version of the experiment has been started. First, I scanned the literature to find a best way how to render the target stimulus unconscious (e.g., through masking). Additionally, literature has also been searched for finding a way how to subjectively measure visual perception of an unconscious target stimulus. That is, a decision has been made together with my supervisor to implement the Perceptual Awareness Scale (PAS, Skóra et al., 2021). Weeks five and six served as programming the new version in PsychoPy. For week seven, the goal was to finetune the masking procedure. That is, several versions of the experiment have been programmed and tried out. By week eight the first version of the unconscious paradigm became ready, thus I performed the full experiment as a first participant. This was then followed by data analysis, which revealed that rendering the target stimulus unconscious was not clearly successful. Results – to be discussed below – indicated that the effects of the conscious paradigm have been vanished.

Method

Participants

I recruited participants through the university, between the ages of 27 and 36, all of them males (the one excluded was female). One of the participants had corrected-to-normal vision. Only one of the participants was left-handed, and at the onset of the experiments only one of them had left eye dominance (myself). As the project was approved by the university and it was considered a pilot study, no consent form or information letter were provided to the participants, in agreement with my supervisor.

Materials

All stimuli were coded in PsychoPy Coder, through Python as the programming language. The design of the stimuli was directly acquired from Carrasco et al. (2004). The fixation point in the center of the screen was a black dot ($0.1^\circ \times 0.1^\circ$), in front of a uniform gray background, and remained visible throughout the experiment. The cue was a small black dot ($0.3^\circ \times 0.3^\circ$), either presented on top of the fixation point (neutral cue), or in the periphery (1.5° above the center, either to the left or to the right of the center, i.e., peripheral cue). The target stimulus was a Gabor patch (sinusoidal grating with a gaussian envelope, in black and white colors), with a size of $2^\circ \times 2^\circ$. Two of these patches were presented on both sides of the fixation point simultaneously, with a 4° eccentricity along the horizontal line. The spatial frequency of both patches was 4 cpd. While Carrasco et al. (2004) also employed 2 cpd spatial frequencies, this was abandoned in our current setup to reach a bigger effect (see Figure 2, panel a in Carrasco et al., 2004). One of the Gabor patches served as the “standard” patch, with a fixed contrast of 6%. The other Gabor patch’s contrast (labelled as the “test” patch) was randomly varied between 2.5% and 16%, with 9 log increments.

In addition to Carrasco et al. (2004), the unconscious version of the experiment employed both a forward and a backward mask. The decision to implement such a “sandwich” design (target being “wrapped around” by masks) was made together with my supervisor, to have the best possible effect of rendering the target unconscious. This stimulus was decided to be a checkerboard pattern, following the guidelines of Smith (2000). The size and location of the masks (two appearing simultaneously from both sides of the fixation point) were the same as with the target patches. The contrast of the mask stimuli was first set to 9%, then increased to 15%. This decision was made based on the practice trials during the finetuning of the masking procedure (week seven, see above). As it turned out, with 9% mask contrasts the target patches were still discernable, in terms of PAS ratings. That is, with lower mask contrasts the amount of *brief glimpse* was higher than the amount of *no experience* (further details on the PAS ratings follows in the next paragraph). With 15% mask contrast, the amount of *brief glimpse* decreased while *no experience* increased (at least, during the finetuning process on week seven).

The unconscious version of the paradigm further utilized PAS ratings on a trial basis, implemented from Skóra et al. (2021). This item asked participants how clear their experience was of the Gabor patches. The rating consists of a scale of four points (1 = *no experience*, 2 = *brief glimpse*, 3 = *almost clear experience*, and 4 = *clear experience*). All texts, including instructions, were presented in black color.

Procedure

Both the original and the unconscious paradigms contained the same number of blocks and trials. Both started with 75 trials as a practice block (including feedback if the participant did not respond), followed by 10 blocks of 198 trials. Since the cue position can either be neutral, peripheral left, or peripheral right, it was decided to reduce the number of trials from the original 200 to 198, thus reaching an equal (but random) amount of cue types (i.e., equal and random conditions). The original experiment lasted about one hour, while the unconscious experiment was about two and a half hours (since PAS ratings followed each trial). Participants were asked to fixate on the fixation point during the entire experiment.

Participants viewed the experiment binocularly on a desktop monitor (1920 x 1080; 60 Hz), from a distance of 96 cm, with a chinrest to exclude head-movements. Light in the room was turned off. The original paradigm followed the same steps as it was reported in Carrasco et al. (2004). The fixation point was first shown for 500 ms, followed by either a neutral or peripheral (right or left) cue, for 67 ms. This was then followed by an interstimulus interval (ISI) of 53 ms, and finally the two Gabor patches were presented for 40 ms. See Figure 1 for further details.

The unconscious paradigm differed in the following way. After the fixation point of 500 ms, the cue was presented for 50 ms. After an ISI of 33 ms, a mask pattern was presented for 33 ms, followed by the target patch for 33 ms, and again a mask pattern was shown for 33 ms. See Figure 2 for further details. Both the original and the unconscious paradigms concluded a trial with a 1000 ms response window. Additionally, after each trial in the unconscious paradigm, the PAS question was presented until the participant responded with one of the keys (“D” – *no experience*, “F” – *brief glimpse*, “J” – *almost clear experience*, “K” – *clear experience*).

Two important notes must be made with respect to the timing of each stimulus. As Carrasco et al. (2004) noted, 120 ms between the cue and target onset has the best effect to have the cue evoking covert transient attention. The authors further remark that the interval between cue onset and target offset should be maximum 160 ms, to exclude eye movements. Accordingly, the time intervals of the unconscious paradigm had to be shortened, which was attained by leveraging the computer’s screen refresh rate (e.g., two frames divided by 60 Hz = 33 ms).

During the 1000 ms response window (both the original and the unconscious experiments), participants had two forced choices to make. They first had to decide on which side from the fixation point the target patch had the higher contrast. More importantly, they were instructed to focus on the direction of the higher contrast target patch. Instruction texts were copied directly from Carrasco et al. (2004). If the target patch on the left was higher in contrast and tilted to the left (by 45°), participants pressed “D”. If it was tilted to the right, participants pressed “F”. If the target patch on the right was higher in contrast and tilted to the left, participants pressed “J”, whereas if it was tilted to the right, they pressed “K” (which is the case on Figure 1 as well). The orientations of the target patches occurred randomly and independently of each other. Participants were also informed that the cue was uninformative either of the contrast or the orientation of the target patches, and that they appeared equally probably. At the end of the procedure, participants received verbal debriefing on the original purpose of the experiment.

Figure 1

(a) Sequence of events in a single trial (directly acquired from Carrasco et al., 2004).
(b) Observers performed a two-by-two forced choice task.

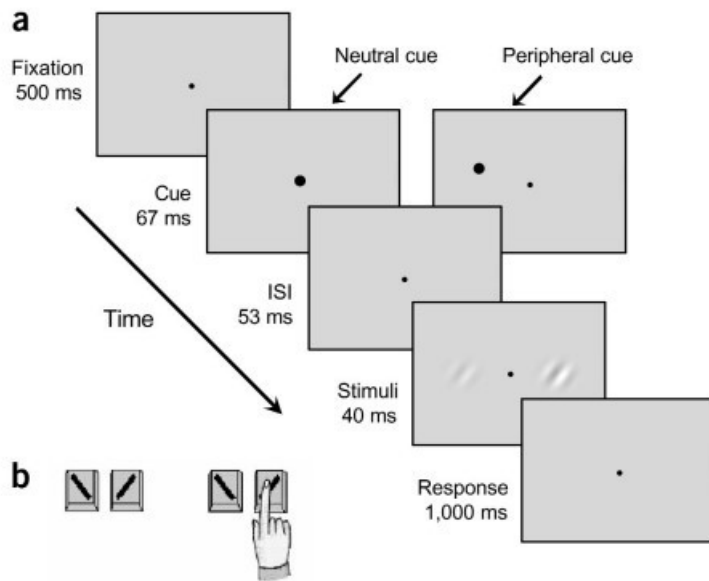
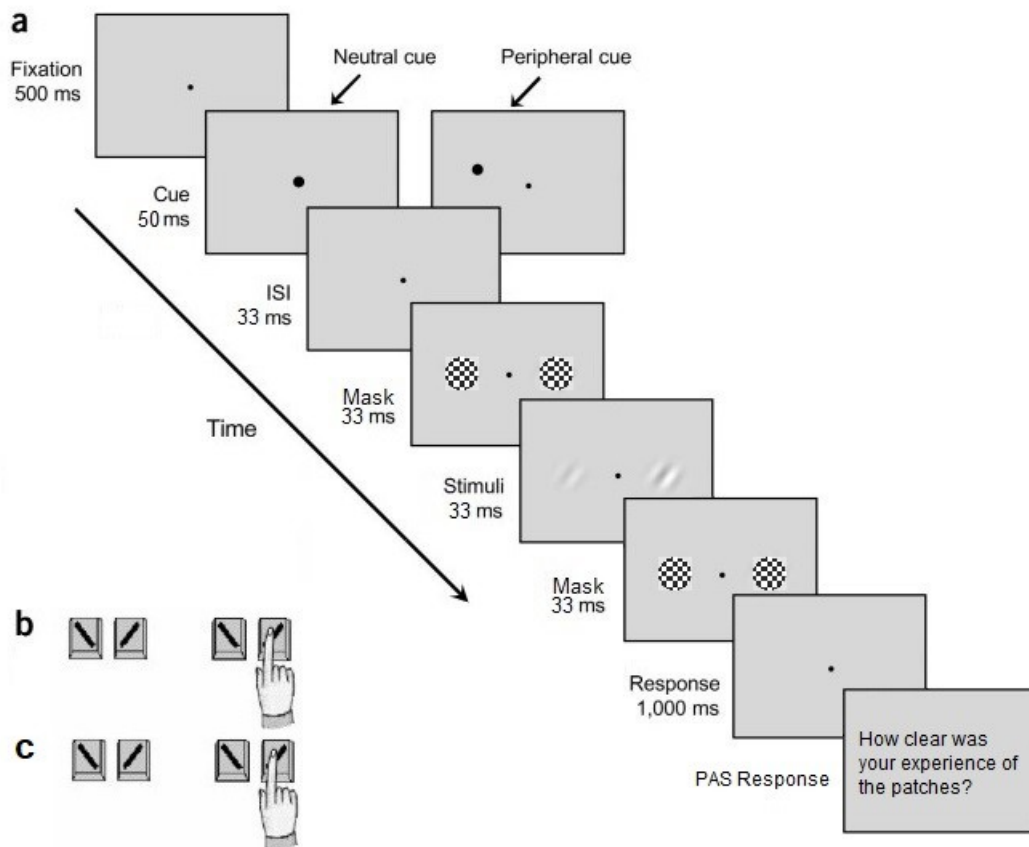


Figure 2

(a) Sequence of events in a single trial (adapted from Carrasco et al., 2004). (b) Observers performed a two-by-two forced choice task. (c) Observers responded to the PAS item.



Results

Data analysis was performed in Microsoft Excel. Figure 3 displays the overall data of five participants, as per the original (conscious) paradigm. Overall accuracy was 63%, meaning that participants were above chance level in responding. Individual data can be further found in the Appendix. On Figure 3, the X axis shows the (objective) contrasts of the “test” patch, whereas the Y axis shows the percentages of (subjective) responses where participants reported the “test” patch being higher in contrast than the “standard” patch. The midline on the Y axis – the 50% – represents the Point of Subjective Equality (PSE). At this point the probability of choosing the “test” patch over the “standard” is 50%, meaning that participants chose either one of the patches with equal probability.

Following the predictions of Carrasco et al. (2004), at the neutral condition (with the neutral cue) the PSE should equal to physical equality. That is, when participants chose either one of the patches with equal probability, the “test” contrast should also equal to the “standard” contrast. This, in fact, was found and can be read of from Figure 3 (blue line). At 50% PSE the objective contrast of the “test” patch is around 6% contrast, being equal to the fixed 6% contrast of the “standard” patch. The neutral cue did not enhance subjective perception of the contrasts.

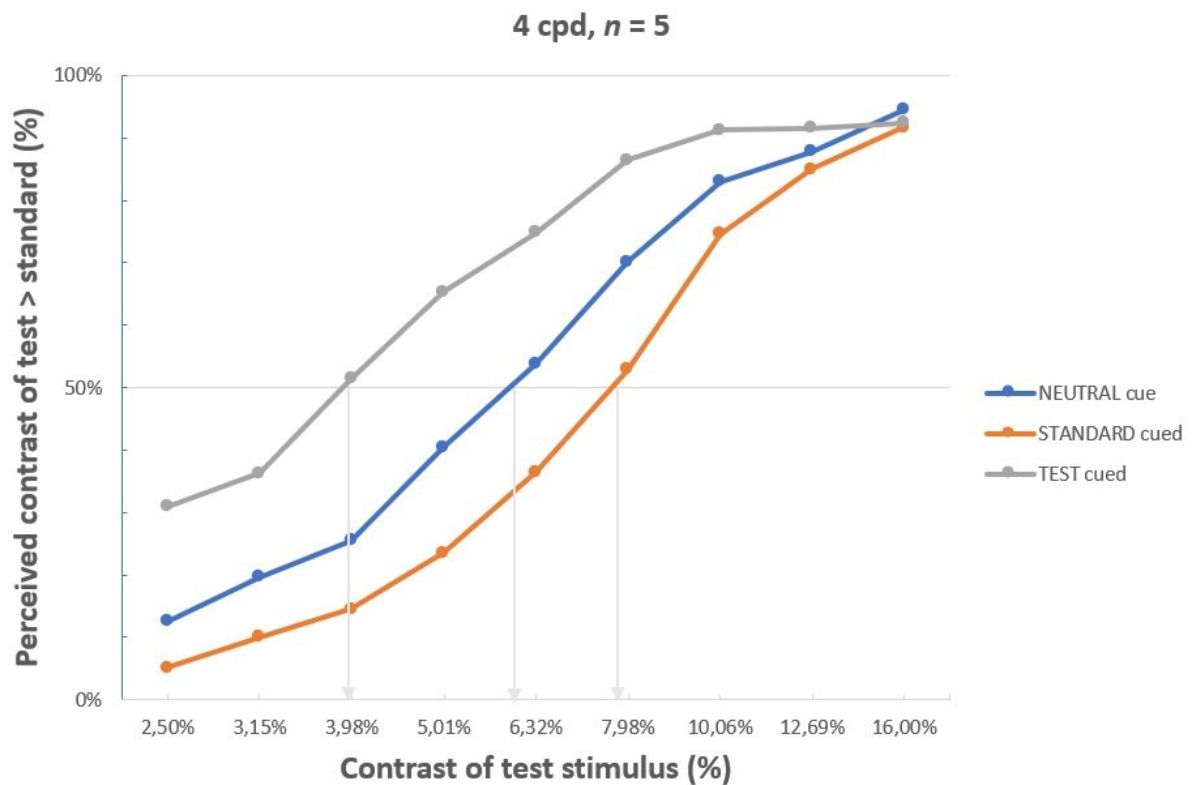
When the “test” patch is cued, it is predicted that the PSE will occur not at objective equality, but at lower “test” contrasts. The reason behind this can be found in the idea that covert transient attention modifies subjective perception of contrasts. When the “test” patch is cued, although it has objectively a lower contrast, participants are expected to rate it as high as the contrast of the “standard” patch. This was, in fact, the case. The grey line of Figure 3 shows that the PSE occurred around 3,98% “test” contrast. Although the contrast of the “test” patch was lower than the “standard” patch, participants were choosing both patches with equal probability.

On the other hand, upon cuing the “standard” patch, we expect higher objective contrasts of the “test” patch for participants to report both patches with equal probability. The orange line on Figure 3 shows this. Although the objective contrast of the “test” patch was around 7,98%, participants still reported subjective equality (PSE).

All together, these results indicate that cuing a target stimulus in the periphery leads to an increased subjective phenomenology of perceived contrasts.

Figure 3

Overall results of five participants. The layout of the graph was adapted from Carrasco et al. (2004).



Next, data was analyzed for the unconscious paradigm. Within this setup, we implemented three conditions (appearing randomly and equally). In the conscious condition the target stimuli were present but not the masks. In the unconscious condition both the target and mask stimuli were present. In the “absent” condition the target was not present, but the masks were. This decision was made so that differences in performance between unconscious and “absent” trials can be verified. The conscious condition further served as baseline trials. The accuracy in the unconscious condition was 55% (including only *no experience* responses from the PAS ratings). This would mean that participants were responding better than chance, that is, the unconscious manipulation was not successful (Rothkirch et al., 2012). Accuracy in the conscious condition was 74% (including only *almost clear experience* and *clear experience* responses from the PAS ratings), while in the “absent” condition it was 50% (including only *no experience* responses from the PAS ratings).

A second point on why the unconscious manipulation was not successful can be made through analyzing the PAS ratings. As mentioned above, not all PAS ratings were utilized, depending on the condition. That is, in the conscious condition only the *almost clear experience* and *clear experience* responses were considered; in the unconscious and “absent” conditions only the *no experience* responses. Looking at the trial level, most responses occurred with the PAS ratings *almost clear experience* and *clear experience*, in the conscious condition. Furthermore, in the “absent” condition most responses occurred with the PAS rating *no experience*. However, in the unconscious condition the amount of *no experience* was lower than *brief glimpse*, suggesting that the unconscious manipulation did not work on the subjective level either. Further numbers are shown on Figure 4.

Figure 4

Amount of PAS ratings (i.e., number of trials), depending on condition.

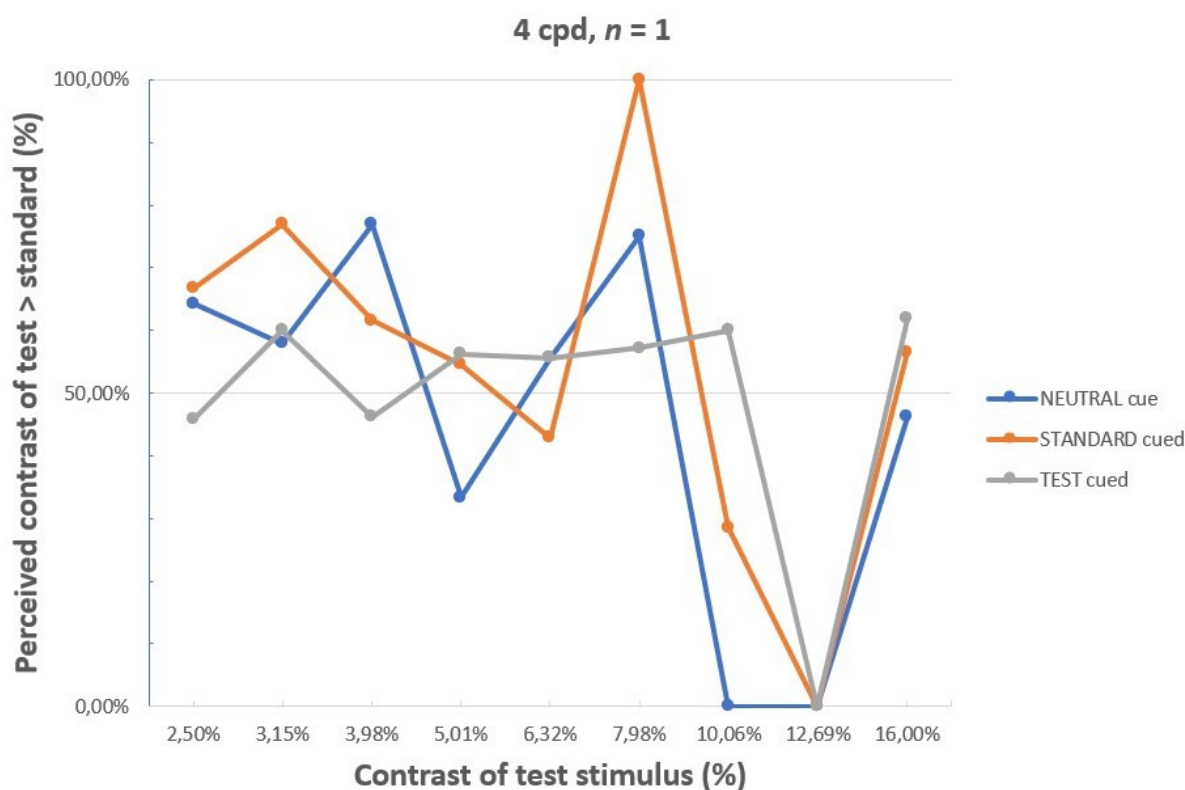
Condition	PAS Rating	Count / PAS Rating
Conscious	1 - no experience	6
	2 - brief glimpse	118
	3 - almost clear experience	477
	4 - clear experience	59
Unconscious	1 - no experience	244
	2 - brief glimpse	409
	3 - almost clear experience	6
	4 - clear experience	1
Absent	1 - no experience	640
	2 - brief glimpse	17
	3 - almost clear experience	3

Although the unconscious manipulation was unsuccessful, we still wanted to see whether our data replicates the conscious paradigm. As such, our predictions were the same as before. In the neutral condition we expected subjective equality to appear at objective equality (i.e., PSE occurring around 6% “test” contrast, being equal to the 6% fixed contrast of the “standard”). In the test condition subjective equality should appear at lower levels of “test” contrasts, and conversely, in the standard condition subjective equality should appear at higher levels of “test” contrasts.

However, these results were not replicated in the unconscious paradigm. Results of one participant – for the unconscious condition – are shown on Figure 5. As the conscious and “absent” conditions similarly did not reveal meaningful results, these figures can be found in the Appendix.

Figure 5

Results of the unconscious paradigm, of one participant, in the unconscious condition.



Discussion

The aim of this project was to see whether covert transient attention intensifies early visual processes, in terms of subjectively perceived contrasts. A secondary goal was to develop a pilot study, to see whether covert transient attention also modifies unconscious visual processing. Our empirical findings give support to the idea that attention can alter appearance (Carrasco et al., 2004). It would be, however, premature to conclude anything about unconscious processes in such subjective perceptions.

Several reasons can be lined up for why the latter might be the case. Smith (2000) for example indicated that when attention is cued, target sensitivity increases with a backward mask. In terms of our unconscious manipulation, this could mean that although our masks suppressed the visual perception of the target stimuli, nonetheless they were still visible or even increased perceived contrast of the target patches. This manifests most clearly in the PAS ratings, where within the unconscious condition more *brief glimpse* ratings were present than *no experience*. This suspicion is reinforced by the theory of reduction of uncertainty (Pelli, 1985, cited in Meese et al., 2007), meaning that a mask gives away information about the target (e.g., orientation), thus reducing uncertainty in the processing of the target stimuli.

Another limitation can be found in the type of the masks. Several studies employed orthogonally oriented Gabor patches, overlayed on the target (e.g., Meese et al., 2007, Freeman & Verghese, 2009, Petrov et al., 2005). In our current setup, however, this was abandoned since participants were asked to respond to the orientation of the target stimuli. That is, an overlayed mask stimulus with an orthogonal orientation might be confusing (or confounding) during the task. Hence, it was decided to use a checkerboard mask, similarly to Smith (2000).

A third limitation, albeit only intuitive, comes from my personal feedback on the unconscious paradigm. While in the conscious paradigm the trials follow in a rapid succession, leading to a sort of “flow” state in the responses, this feeling was vanished in the unconscious paradigm, where after each response one also had to make a choice on the PAS rating. This also produced a prolonged process (about two and a half hours), that resulted in me becoming tired but also inducing afterimages quite often, meaning that my vision became unreliable after a while. A possible solution would be to reduce the number of trials, or perhaps exclude PAS ratings on a trial level.

There is an additional concern that was left out of the scope of the current project. For example, it is unclear whether eyesight, handedness, or even eye dominance has an influence in these cognitive processes. Could it be the case that right-handed participants responded more with their right hands? Or perhaps a right-eye dominance resulted in focusing more on the right-side of the screen? These questions have not been addressed in this project, although data is available for future analysis.

As for suggestions for future studies, the following points can be maintained. Our unconscious paradigm employed masks with the same size as the targets, overlayed on one another. Petrov et al. (2005), however, reported that surround masks (a ring with a whole inside where the target would appear) in the periphery can suppress visibility of the target (in case the orientations of the target and mask are the same). Meese et al. (2007) also used surround masks (in addition to the aforementioned orthogonal masks). However, caution must be taken upon implementing a surround mask in our unconscious paradigm. As the peripheral cues are placed above the targets, a mask that is bigger in size than the target might reinforce (or vanish) the effects of covert transient attention. This could be circumvented by properly calibrating the location of the cues (e.g., by placing them in a higher position) and the size of the rings of the masks. A similar suggestion can be made upon simply increasing the size of the masks (without an internal ring), which is referred to as *doughnut masks* in Meese et al. (2007).

As a parallel enterprise, one might be interested in not the effects of covert transient attention on unconscious processing of perceived contrasts, but on the effects of unconscious covert transient attention on appearance. This idea came up during the literature

review (weeks four and five, during data collection). McCormick (1997) reported that an exogenous cue can influence attention even without awareness. His study showed that by rendering the cue barely visible (dark grey cue on a black background), participants reaction times were faster to the subsequent target than with a conscious cue (in white color). This effect was reinforced by subjective reports of the participants not being aware of the unconscious cues. Such a design could easily be implemented in the current paradigm in terms of, for example, presenting the cues in light grey color (on the existing uniform grey background). Furthermore, our experimental design also measured reaction times, which was not analyzed. Such an approach could offer an additional opportunity for future studies. It could be the case that subjective appearance can be altered by unconscious attention.

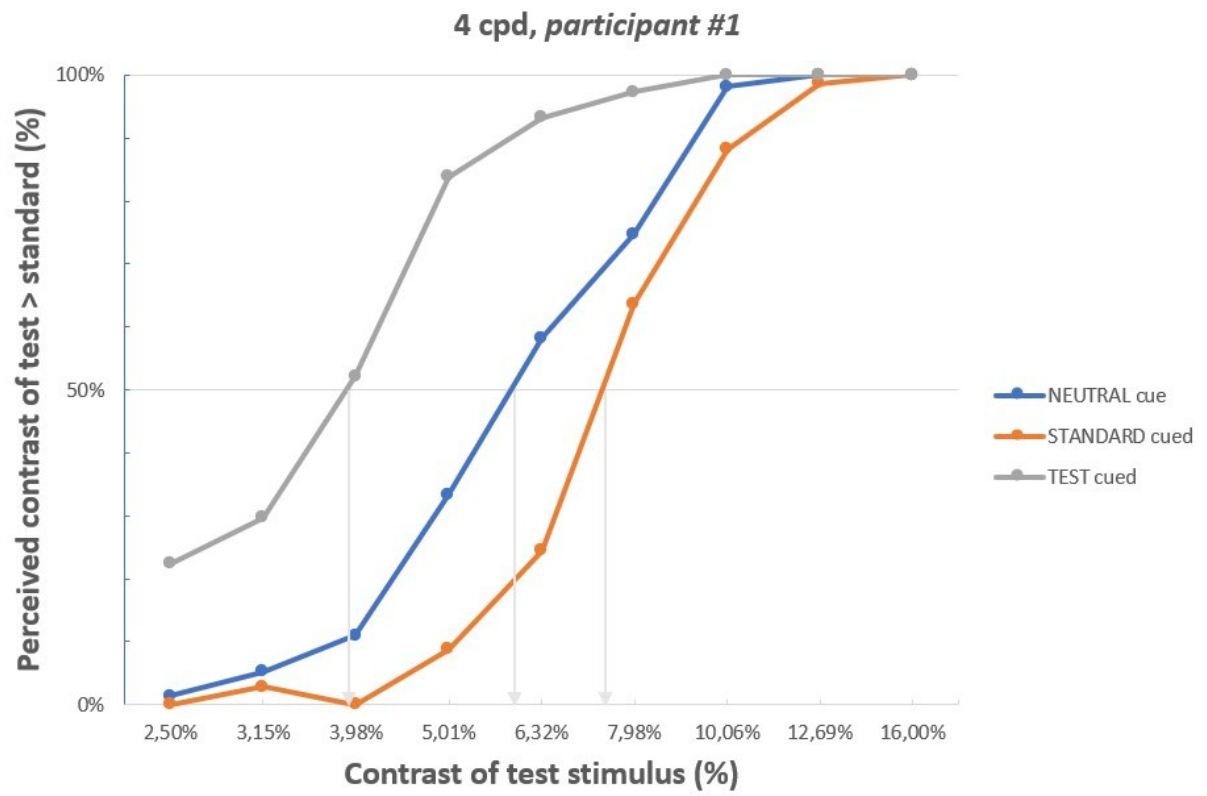
Altogether, it can be claimed that this individual project was successful, in terms of the planned deliveries (more on that in the self-reflection report). The primary aim, to replicate the experimental paradigm of Carrasco et al. (2004), was achieved. Although the unconscious paradigm requires further finetuning, the process of creating such a design offered unique insights into how early visual processing might work. Every day we face a multitude of external stimuli, whereby our attentional processes help to filter out irrelevant information. Even without directing our gaze to such stimuli, our visual perception of the periphery seems to be heightened by such cognitive processes. Thus, our attentional processes become a key factor in our overall cognitive system.

References

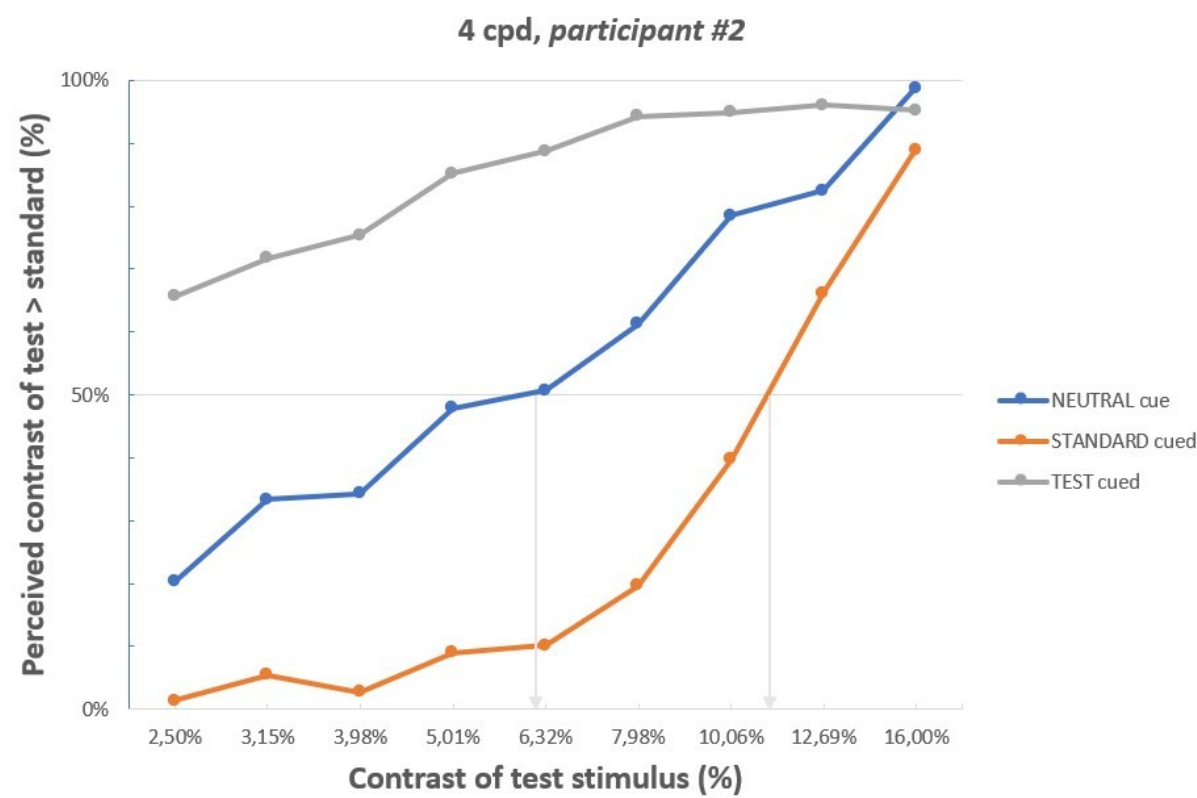
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Appendix

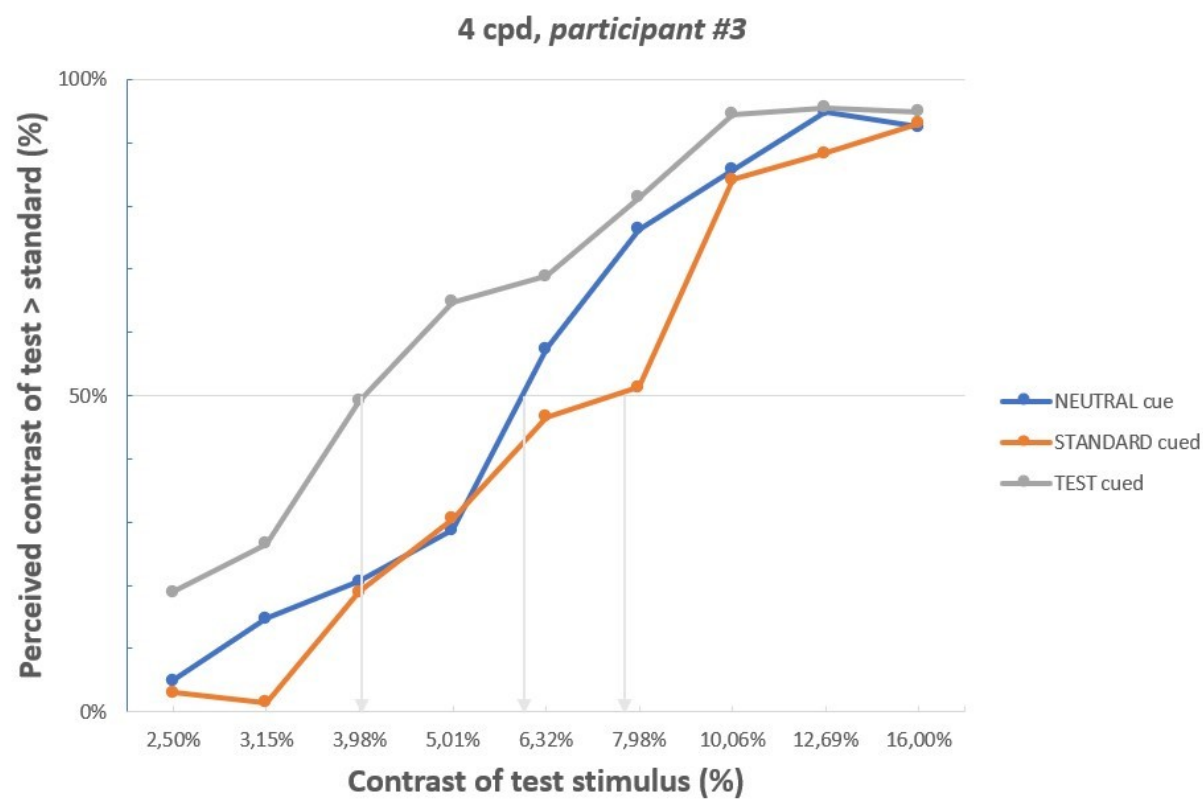
Results of the First Participant, Conscious Paradigm



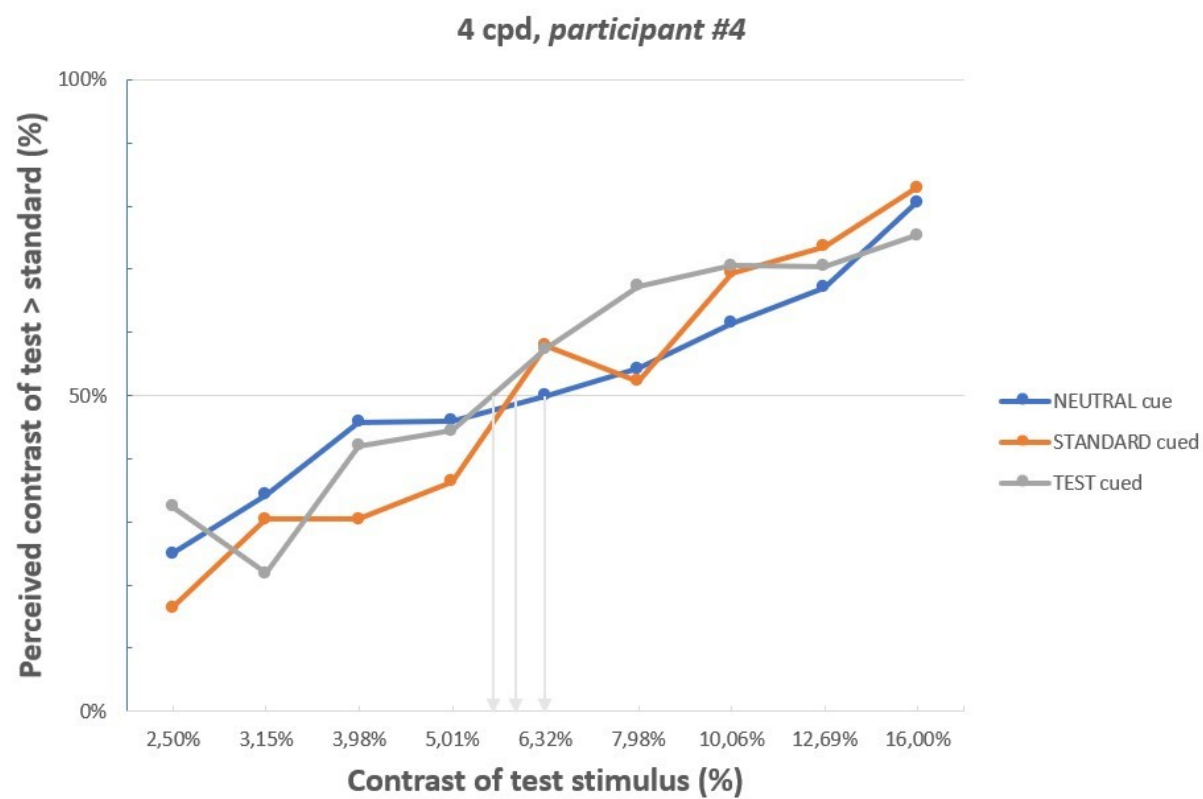
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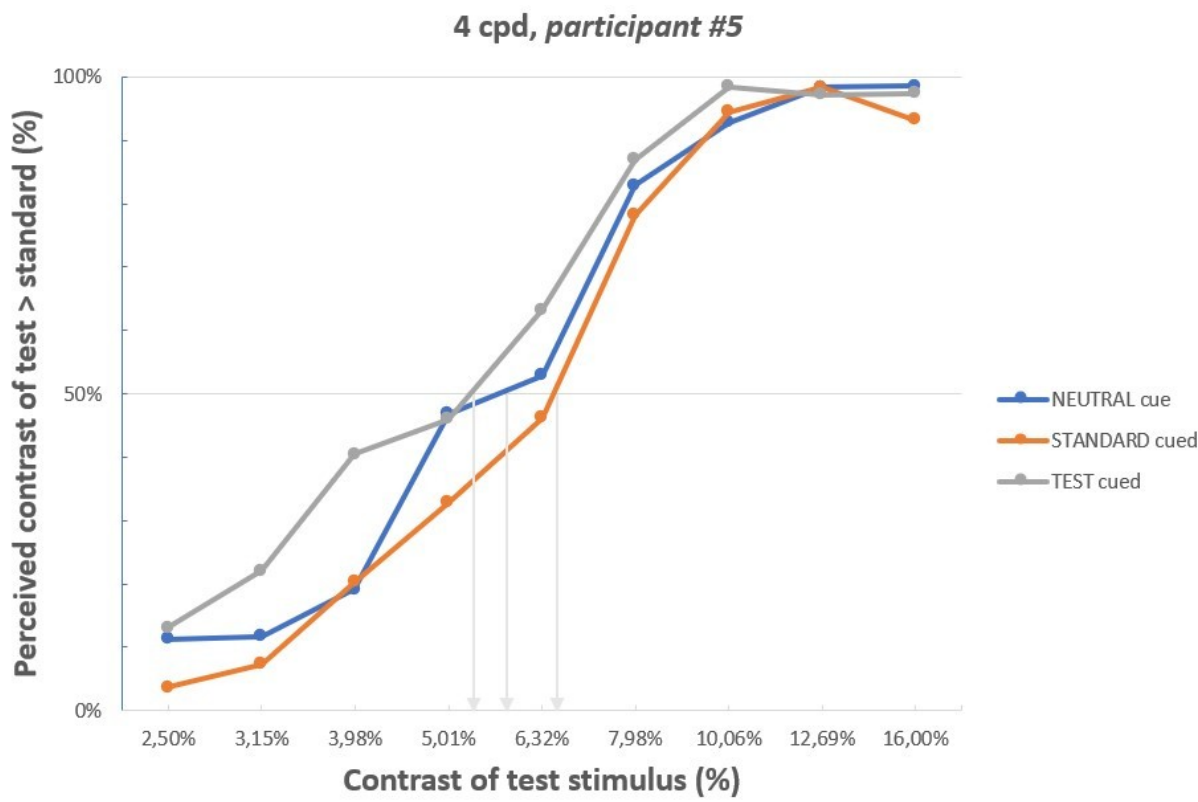
Results of the Third Participant, Conscious Paradigm



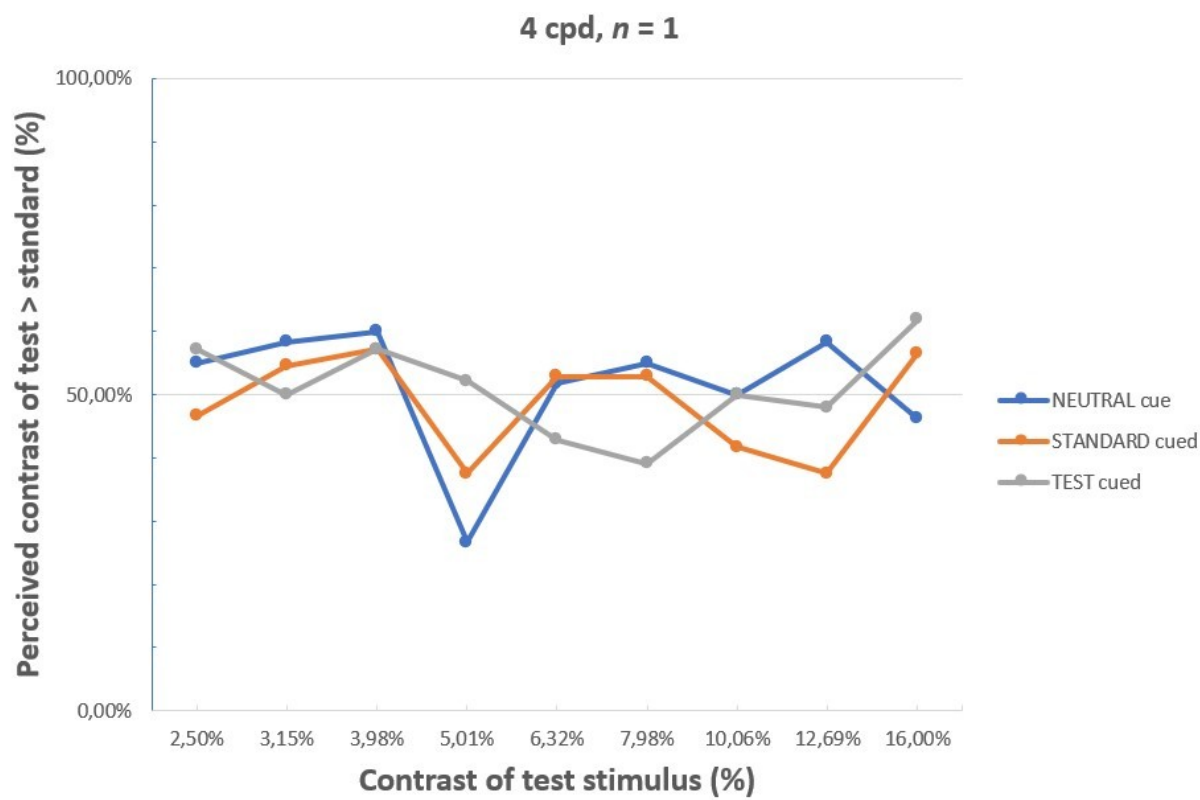
Results of the Fourth Participant, Conscious Paradigm



Results of the Fifth Participant, Conscious Paradigm



Results of One Participant, Unconscious Paradigm, Conscious Condition



Results of One Participant, Unconscious Paradigm, “Absent” Condition

